Resonant no-go for the infinitesimal Class-II tilted commutator bound

TECT verification-first repository · claim A9-CLASSII-SMART-PATH-CANCELLATION

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1. Purpose and scope

Exact result. The commutator-alone estimate designated `A7-CLASSII-TILTED-COMMUTATOR-FORM-BOUND`, with the same arbitrarily small coefficient multiplying entropy and the sextic norm, is false. A physical scalar ray and a same-shell resonant Fourier triad give an explicit positive lower threshold for that coefficient.

Boundary. This does not withdraw the A9 scoped T5 smart-path and frozen-shell theorem, and it neither proves nor disproves the full A7 Nelson bound. The counterexample identifies the positive source part of the covariance-normal frozen energy that the next sufficient estimate may use.

The authority is the hash-pinned A1 production functional on $\mathbb{T}_L^3$, $L = 16$, with $Y = 1$, fixed $\varepsilon_\rho = 10^{-12}$, and

$$a = \frac{c_{JJ}\alpha_X^2}{M_X^2 + \delta_M} = 0.0045.$$

All three nonconstant frequencies below belong to one dyadic annulus. Products are exact Fourier products; the independent audit also computes the embedded Pauli currents directly. For the counterexample sequence we pin the sharp radial filtration

$$P_{\le j-1} = \mathbf{1}_{\{|q|<K\}}, \qquad P_{\le j} = \mathbf{1}_{\{|q|<2K\}}. \tag{1.1}$$

1 Exact scalar-ray reduction

Take a real field in the first internal component, $\Psi = fe_1$. Only generator T_3 contributes, and

$$J_3 = 2f\nabla f, \qquad K_3 = \frac{\varepsilon_\rho}{f^2 + \varepsilon_\rho} J_3. \tag{2.1}$$

Consequently the coefficient in the e_1 derivative direction is

$$B_{11}(fe_1) = 4f^2 \left[a + 2b\frac{\varepsilon_\rho}{f^2 + \varepsilon_\rho} + c\left(\frac{\varepsilon_\rho}{f^2 + \varepsilon_\rho}\right)^2 \right]. \tag{2.2}$$

For fields uniformly separated from zero this is $4af^2 + O(1)$, and the b, c and floor terms do not contribute to the leading high-frequency coefficient.

2 Same-shell resonant triad

Let

$$g_K(x,y) = \cos(Kx) + \cos(Ky) - \cos(K(x+y)). \tag{3.1}$$

Its frequencies have lengths $K, K, \sqrt{2}K$, so they lie in the annulus $[K, 2K)$. Exact Fourier convolution gives

$$\langle g_K |\nabla g_K|^2\rangle = -K^2, \qquad \langle g_K^2 |\nabla g_K|^2\rangle = \frac{5}{2}K^2, \tag{3.2}$$

and

$$\langle g^m\rangle_{m=0}^6 = \left(1, 0, \frac{3}{2}, -\frac{3}{2}, \frac{45}{8}, -\frac{45}{4}, \frac{255}{8}\right). \tag{3.3}$$

Choose the previous shell to be $\phi_{j-1} = Ae_1$ and the full shell to be

$$\phi_j = A(1+\epsilon g_K)e_1, \qquad 0 < \epsilon < \frac{1}{3}. \tag{3.4}$$

This restriction keeps the field uniformly away from zero. Substitution of (3.2) into the raw coefficient increment gives

$$\frac{1}{L^3}\mathcal{C}_j^{\mathrm{raw}} = -aA^4K^2\epsilon^3(4-5\epsilon) + o(A^4K^2). \tag{3.5}$$

The covariance counterterm grows only as $\Gamma_{\le j}O(A^2) = O(K^3)$ when $A = tK$. Thus it cannot cancel the K^6 term in (3.5). Fixed mass and vacuum terms are still lower order.

3 Covariance-contracted Gaussian contradiction

Let C_J be the cutoff covariance, let $d_J = O(K^3)$ be its real dimension, and put

$$h_K = tK(1+\epsilon g_K)e_1, \qquad \sigma_K = K^{-2}, \qquad \nu_K = N(h_K, \sigma_K^2 C_J). \tag{4.1}$$

This law is absolutely continuous with respect to $\gamma_J = N(0, C_J)$ at every finite cutoff. Its exact relative entropy is

$$H(\nu_K \mid \gamma_J) = \frac{1}{2}\|h_K\|_{\mathcal{H}_J}^2 + \frac{d_J}{2}(\sigma_K^2 - 1 - \log\sigma_K^2). \tag{4.2}$$

The lattice-shell count in dimension three and the production q^4 symbol give, uniformly up to the cutoff,

$$\sum_{|q|\le K}\frac{1}{1+|q|^4} = O(1), \qquad \sum_{|q|\le K}\frac{|q|^2}{1+|q|^4} = O(K), \qquad d_J = O(K^3). \tag{4.2a}$$

These follow by grouping lattice points into unit radial shells, whose cardinality is $O(r^2)$, and comparing the resulting sums with integrals. The second term in (4.2) is therefore $O(K^3\log K) = o(K^6)$. The production symbol and (3.3) give

$$\frac{H(\nu_K \mid \gamma_J)}{L^3K^6} \longrightarrow c_H t^2, \qquad c_H = \frac{3}{2}Y\epsilon^2, \tag{4.3}$$

$$\frac{\mathbb{E}_{\nu_K}\|\phi_J\|_6^6}{L^3K^6} \longrightarrow c_6 t^6, \qquad c_6 = M_6(\epsilon) = \langle(1+\epsilon g)^6\rangle, \tag{4.4}$$

$$\frac{\mathbb{E}_{\nu_K}\sum_{\ell\le J}\mathcal{C}_\ell}{L^3K^6} \longrightarrow -c_C t^4, \qquad c_C = a\epsilon^3(4-5\epsilon) > 0. \tag{4.5}$$

Here is the fluctuation estimate explicitly. At fixed floor, B is smooth with quadratic growth and its first two derivatives have polynomial growth. Before contraction the point-value variance is $O(1)$ and the point-derivative variance is $O(K)$; after multiplication by $\sigma_K^2 = K^{-4}$ they are $O(K^{-4})$ and $O(K^{-3})$. Hence the value and derivative standard deviations are $O(K^{-2})$ and $O(K^{-3/2})$. Against $|h_K| = O(K)$ and $|\nabla h_K| = O(K^2)$, the largest Taylor/Wick corrections to the raw term are respectively

$$O(K^{-1})O(K^4) = O(K^3), \quad O(K^2)O(K^2)O(K^{-3/2}) = O(K^{5/2}), \quad O(K^2)O(K^{-3}) = O(K^{-1}). \tag{4.5a}$$

Second-order coefficient corrections are $O(1)$ after multiplication by $|\nabla h_K|^2$. The same moment accounting bounds every lower-shell term by $O(K^3)$, so the $O(\log K)$ shell sum is still $o(K^6)$. The covariance trace subtraction is $O(K^3)$, and all sextic fluctuation corrections are at most $O(K^3)$. Fixed-floor Gaussian tail bounds justify the Taylor remainder. This proves (4.4)–(4.5), rather than assuming that a same-covariance translation is already deterministic.

The falsified estimate would therefore require, for all $t > 0$,

$$c_C t^4 \le \eta(c_H t^2 + c_6 t^6). \tag{4.6}$$

Optimising in t yields the necessary condition

$$\eta \ge \eta_{\min} = \frac{c_C}{2\sqrt{c_H c_6}} > 0, \qquad t^4 = \frac{c_H}{c_6}. \tag{4.7}$$

This contradicts the quantifier *for every* $\eta > 0$. More generally, separate entropy and sextic budgets α, ϵ_6 must obey

$$\alpha\epsilon_6 \ge \frac{c_C^2}{4c_H c_6}. \tag{4.8}$$

At the production point choose $\epsilon = 3/10$. Then

$$M_6 = \frac{4412239}{1600000} = 2.757649375, \qquad \eta_{\min} = 2.4891432\,10^{-4}. \tag{4.9}$$

Hence $\eta = 10^{-4}$ gives a strict K^6 violation; no fixed C_η can repair it as $K \to \infty$. The fixed quartic term is only $O(K^4)$ and likewise cannot change the contradiction.

4 Why the positive A9 theorem survives

The covariance-normal frozen shell term contains a positive source energy whose leading coefficient on the same ray is

$$\frac{Q_{\mathrm{fr}}^{\mathrm{source}}}{L^3K^6} \longrightarrow c_F t^4, \qquad c_F = 4a\epsilon^2. \tag{5.1}$$

The ratio forced by the witness is

$$\theta_{\mathrm{ray}} = \frac{c_C}{c_F} = \frac{\epsilon(4-5\epsilon)}{4}. \tag{5.2}$$

For $\epsilon = 3/10$, $\theta_{\mathrm{ray}} = 3/16 = 0.1875$. Thus the commutator is compensated by a fixed part of the frozen energy. A9 proved the exact determinant and entropy control for the complete covariance-normal frozen term; it never proved the now-falsified commutator-alone estimate.

5 Corrected open gate

Let Q_j^{fr} denote the complete covariance-normal frozen shell energy, including its trace subtraction. The corrected gate is

$$\boxed{\texttt{A7-CLASSII-FROZEN-ENERGY-RELATIVE-COMMUTATOR-BOUND.}}$$

For some fixed $\theta \in (0,1)$ set

$$R_j^\theta = \theta Q_j^{\mathrm{fr}} + \mathcal{C}_j. \tag{6.1}$$

The target is to prove explicit production-budget constants $\alpha_c, \epsilon_6, K_\theta, C_\theta$ such that, uniformly in cutoff and all $\nu \ll \gamma_J$,

$$\mathbb{E}_\nu \sum_{j\le J} R_j^\theta \ge -\alpha_c H(\nu \mid \gamma_J) - \epsilon_6 \mathbb{E}_\nu \|\phi_J\|_6^6 - K_\theta \mathbb{E}_\nu \|\phi_J\|_4^4 - C_\theta. \tag{6.2}$$

The untouched $(1-\theta)Q_j^{\mathrm{fr}}$ remains under A9's $\det_2$/Schatten theorem. A closure must use the full covariance-normal quantity, not only its raw positive square; otherwise an uncancelled Schatten-one trace is reintroduced. On the registered ray the exact necessary tradeoff is

$$\alpha_c \epsilon_6 \ge \frac{[(c_C - \theta c_F)_+]^2}{4 c_H c_6}. \tag{6.3}$$

Thus $\theta = 3/16$ neutralises this ray without entropy or sextic expenditure; it is not an absolute lower bound on θ when positive budgets are allowed. This package proves the necessary resonant tradeoff and records a candidate closure target; it does not prove (6.2) for all fields.

6 Devil's-advocate review

1. **Objection: a single Fourier mode has no negative term. DISMISSED.** That mode misses the exact triad resonance (3.2).
2. **Objection: the floor or K_A term repairs the sign. DISMISSED.** The witness stays uniformly away from zero, and (2.2) shows that b, c, ε_ρ are lower order.
3. **Objection: the covariance counterterm cancels the defect. DISMISSED.** It is $O(K^3)$ while the defect is $O(K^6)$.
4. **Objection: the tilted law is not admissible. DISMISSED.** It is a finite-cutoff nondegenerate Gaussian with a Cameron–Martin mean and contracted positive covariance. It is absolutely continuous with respect to the reference Gaussian and has the exact finite entropy (4.2).
5. **Objection: this disproves the full Nelson estimate. UPHELD as false.** Equation (5.1) supplies a positive term discarded by the failed decomposition.
6. **Objection: a raw-square split is sufficient. UPHELD as invalid.** The fraction must be applied to the complete covariance-normal frozen energy or the trace divergence returns.
7. **Objection: the corrected relative bound is now proved. UPHELD as an overclaim.** Only its necessary resonance budget and a candidate closure target are recorded; the all-field estimate is open.
8. **Objection: every corrected proof requires $\theta \ge 3/16$. UPHELD as false.** That value makes the registered ray cost-free; smaller values remain possible if the explicit budget tradeoff (6.3) is satisfied.

7 Reproduction and references

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python codes/foundations/a9_tilted_commutator_nogo_verify.py
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The primary route uses exact rational Fourier convolution and the independent route evaluates the embedded Pauli currents without importing the primary code. The variational-law background follows M. Boue and P. Dupuis, https://doi.org/10.1214/aop/1022855876. The constructive variational context is related to N. Barashkov and M. Gubinelli, https://arxiv.org/abs/1805.10814. Neither reference supplies the model-specific no-go result proved here.

Result footer

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Claim: A9-CLASSII-SMART-PATH-CANCELLATION
Result: A7-CLASSII-TILTED-COMMUTATOR-FORM-BOUND falsified as stated
Preserved: A9 scoped T5 exact smart-path and frozen-shell theorem
Open: A7-CLASSII-FROZEN-ENERGY-RELATIVE-COMMUTATOR-BOUND
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Final boundary. The infinitesimal commutator-alone sufficient condition is eliminated. The A9 T5 theorem remains valid. The self-coupled Nelson estimate, the corrected all-field relative bound, the interacting Gibbs measure, floor removal, infinite volume, phase transition, BCC selection, T6, and T7 are not claimed.